Single generators of polynomial rings under composition: a characteristic-two dichotomy

— condensed version —

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Abstract

Say that $p \in R[x, y]$ *generates* $R[x, y]$ if every bivariate polynomial over R is an iterated composition of p, the variables, and constants. We prove: there is a single polynomial generating all of $R[x, y]$ under composition if and only if 2 is a unit of R; when it is, $x^2 - y$ is a generator, and when it is not, for every p at least one of $x + y$, xy is inexpressible.

1 Definitions and statements

Definition 1. Let R be a commutative ring, $R[x, y]$ the ring of formal polynomials in two variables, and $p \in R[x, y]$. The set $\mathrm{Clo}(p) \subseteq R[x, y]$ is the smallest set containing x, y, and every constant $c \in R$, and closed under $(f, g) \mapsto p(f, g)$ (substitution of f for x and g for y in p). Say p *generates* $R[x, y]$ if $\mathrm{Clo}(p) = R[x, y]$.

In clone-theoretic terms: the binary terms in the language of commutative rings with constant symbols form the binary part of an abstract clone, canonically identified with $R[x, y]$ under substitution, and our question asks when the single term p generates it. This abstract object is represented faithfully by operations on R when R is an infinite integral domain, and is finer than its image otherwise; it should not be confused with the clone of polynomial operations on R in the sense of universal algebra, nor with the full clone of all finitary operations on R. The distinction between formal polynomials and the functions they induce matters over finite fields, and the contrast is sharpest there: by a theorem of Webb [5], a single binary operation generates every operation on any finite set — in particular every polynomial function over $\mathbb{F}_2$ is a composition of a single binary polynomial function — while Theorem 3 shows that no single polynomial generates $\mathbb{F}_2[x, y]$ formally. The formal statement therefore has no functional counterpart. Concretely, $n := 1 + xy \in \mathbb{F}_2[x, y]$ induces the binary NAND operation on $\mathbb{F}_2$ (the Sheffer stroke [4]), and every Boolean function is a composition of NAND; nevertheless $x + y \notin \mathrm{Clo}(n)$ by Theorem 3, since $Dn = 1 \neq 0$ for the operator D of Section 3.

Theorem 1 (dichotomy)**.** *Let R be a commutative ring. There is a single polynomial generating all of $R[x, y]$ under composition (with variables and constants) if and only if 2 is a unit of R.*

Theorem 2. *If $\frac{1}{2} \in R$ then $q(x,y) := x^2 - y$ generates the full polynomial ring:* $\mathrm{Clo}(x^2 - y) = R[x,y]$.

Theorem 3. *For every $q \in \mathbb{F}_2[x,y]$, $\ x + y \notin \mathrm{Clo}(q)$ or $xy \notin \mathrm{Clo}(q)$.*

There is no hypothesis on q: reduction mod 2 collapses degree ($2x^3 + xy \mapsto xy$), so the deduction of Corollary 5 below requires Theorem 3 for arbitrary q, degenerate cases included.

Theorem 4. *For every field k of characteristic 2 and every $q \in k[x,y]$, at least one of $x+y$, xy lies outside $\mathrm{Clo}(q)$.*

Corollary 5. *No binary polynomial generates $\mathbb{Z}[x,y]$.*

Lemma 6. $\mathrm{Clo}(p) = R[x,y]$ *if and only if $x + y \in \mathrm{Clo}(p)$ and $xy \in \mathrm{Clo}(p)$.*

Proof. Substituting members of $\mathrm{Clo}(p)$ into a member lands in $\mathrm{Clo}(p)$ (induction on the formula). So if $x+y, xy$ are members, $\mathrm{Clo}(p)$ is closed under $+$ and $\times$; containing the variables and constants, it is everything. □

Lemma 7 (transfer)**.** *A ring homomorphism $R \to S$ induces $\pi : R[x,y] \to S[x,y]$ with $\pi\big(\mathrm{Clo}(p)\big) \subseteq \mathrm{Clo}(\pi p)$.*

Proof. π fixes variables, sends constants to constants, and commutes with substitution; induct over the formula. □

By Lemmas 6 and 7 applied to $\mathbb{Z} \to \mathbb{F}_2$, Theorem 3 implies Corollary 5.

We use three facts about the formal partials ∂_x, ∂_y on $\mathbb{F}_2[x,y]$: (1) $\partial(f^2) = 2f\,\partial f = 0$; (2) $\partial_x^2 = \partial_y^2 = 0$, while $D := \partial_x \partial_y$ need not vanish; (3) squaring is a ring homomorphism, and a polynomial with all exponents even is a square. Univariate degrees follow the convention that constants — including 0 — have degree 0, so $\deg(f^2) = 2\deg f$ without exception.

Related work. Webb's theorem [5] settles the function-setting analogue on every finite set in the positive direction, in contrast with Theorem 3. The nearest prior work on formal polynomials we know of is that of Aichinger and Kreinecker on generation in $(\mathbb{Z}[x], +, \circ)$ [1, 2]: univariate, over $\mathbb{Z}$, with addition retained as a primitive, and including 2-divisibility obstructions such as $2x^{10} \in \langle x^2 \rangle$ but $x^{10} \notin \langle x^2 \rangle$. We are not aware of prior work on the composition-only single-generator question for formal polynomial rings.

2 The positive half: $x^2 - y$

Proof of Theorem 2. By Lemma 6 it suffices to reach $x + y$ and xy. All steps below are instances of

$$q\big(\alpha,\, q(\alpha', \gamma)\big) = \alpha^2 - \alpha'^2 + \gamma. \tag{1}$$

Translations. With constants $\alpha = \frac{d+1}{2}$, $\alpha' = \frac{d-1}{2}$, (1) sends $\gamma \mapsto \gamma + d$ for any $d \in R$ (the only step using $\frac{1}{2}$); in particular $x + d, y + d \in \mathrm{Clo}(q)$.

Slanted lines. $q(c,y) = c^2 - y$, then $q(x, c^2 - y) = x^2 + y - c^2$, then

$$q\big(x + d,\ x^2 + y - c^2\big) = 2dx - y + (d^2 + c^2) \ \in\ \mathrm{Clo}(q),$$

a line of slope $2d$ for every d, constant adjustable by translations.

Linear span step. For $\alpha \in \mathrm{Clo}(q)$ and constant k, (1) with outer argument $\alpha + k$ and inner argument α gives the increment

$$\gamma \longmapsto \gamma + 2k\,\alpha + k^2.$$

Increments along slanted lines of two distinct slopes span all linear polynomials, so from $\gamma = 0$: every linear polynomial lies in $\mathrm{Clo}(q)$, in particular $x + y$ and $x + \frac{y}{2}$.

Scaled squares. The increment step with $\alpha = y^2 + \ell$ (in the clone via $q(y, -\ell)$) adds $\lambda y^2 + \text{const}$ for any λ; from $\gamma = x^2 = q(x, 0)$ we get $x^2 + \frac{1}{4}y^2 \in \mathrm{Clo}(q)$.

Finish.

$$q\Big(x + \tfrac{y}{2},\ x^2 + \tfrac{1}{4}y^2\Big) = \Big(x + \tfrac{y}{2}\Big)^2 - x^2 - \tfrac{1}{4}y^2 = xy. \qquad \square$$

3 The negative half over $\mathbb{F}_2$

Fix $q \in \mathbb{F}_2[x, y]$. Grouping the monomials of q by the parities of their exponents and using fact (3), q has a unique *parity decomposition*

$$q \;=\; A_0^2 \;+\; xA_1^2 \;+\; yA_2^2 \;+\; xy\,A_3^2, \qquad A_i \in \mathbb{F}_2[x, y], \tag{2}$$

and differentiating, with squares derivative-free,

$$\partial_x q = A_1^2 + yA_3^2, \qquad \partial_y q = A_2^2 + xA_3^2, \qquad Dq = A_3^2. \tag{3}$$

Lemma 8 (chain rule for D)**.** *In characteristic* 2*, for all* $q, \alpha, \beta \in \mathbb{F}_2[x, y]$*,*

$$D\big(q(\alpha, \beta)\big) = (Dq)(\alpha, \beta)\,\big(\partial_x\alpha\,\partial_y\beta + \partial_y\alpha\,\partial_x\beta\big) + (\partial_x q)(\alpha, \beta)\,D\alpha + (\partial_y q)(\alpha, \beta)\,D\beta.$$

Proof. Expand $\partial_x\partial_y\big(q(\alpha, \beta)\big)$ by the chain and product rules; the terms involving $\partial_x^2 q$ or $\partial_y^2 q$ vanish by fact (2), and the survivors assemble as displayed. $\square$

Proposition 9. *If* $Dq = 0$ *then* $xy \notin \mathrm{Clo}(q)$*.*

Proof. $\{g : Dg = 0\}$ contains the variables and constants and, by Lemma 8 with the first term dead, is closed under substitution into q; hence it contains $\mathrm{Clo}(q)$. But $D(xy) = 1$. $\square$

Suppose henceforth $Dq \neq 0$, i.e. $A_3 \neq 0$; we show $x + y \notin \mathrm{Clo}(q)$. Write $\mathbb{F}_2[x{+}y]$ for the subring of polynomials in $x + y$. The proof runs through the rigidity property

> (†) for all $\alpha, \beta \in \mathbb{F}_2[x, y]$: whenever $q(\alpha, \beta)$ is a nonconstant element of $\mathbb{F}_2[x{+}y]$, both $\alpha, \beta \in \mathbb{F}_2[x{+}y]$.

Proposition 10 (from (†) to non-generation)**.** *If q satisfies* (†) *then* $x + y \notin \mathrm{Clo}(q)$*.*

Proof. We claim every $g \in \mathrm{Clo}(q)$ satisfies $P(g):\ g \notin \mathbb{F}_2[x{+}y]$ or g constant. The substitution $x \leftrightarrow y$ fixes $\mathbb{F}_2[x{+}y]$ pointwise but moves x and y, so the variables satisfy P; constants do. Let $g = q(\alpha, \beta)$ with $P(\alpha), P(\beta)$ and suppose $g \in \mathbb{F}_2[x{+}y]$ is nonconstant. Then (†) forces $\alpha, \beta \in \mathbb{F}_2[x{+}y]$; P forces them constant; then g is constant — contradiction. So P holds on $\mathrm{Clo}(q)$, and $x + y$ fails P. $\square$

Fix an algebraically closed field $K \supseteq \mathbb{F}_2$ of characteristic 2 containing an element σ transcendental over $\mathbb{F}_2$; concretely $K = \overline{\mathbb{F}_2(\sigma)}$. Call $c \in K$ *transcendental* if it satisfies no nonzero polynomial relation over $\mathbb{F}_2$.

Definition 2. A *witness* is a pair $\alpha, \beta \in K[t]$, not both constant, with $q(\alpha, \beta) = c$ constant and c transcendental.

Proposition 11 (from curves to (†))**.** *If no witness exists, then q satisfies (†).*

Proof. Let $q(\alpha, \beta) = f(x + y)$ with $f \in \mathbb{F}_2[u]$ nonconstant. Substitute $(x, y) = (t,\, t + \sigma)$: the right side becomes the constant $f(\sigma)$, which is transcendental (a nonzero $\mathbb{F}_2$-relation on $f(\sigma)$ composes with f to one on σ). Since no witness exists, $\alpha(t, t{+}\sigma)$ and $\beta(t, t{+}\sigma)$ are constant in t. Finally, a $g \in \mathbb{F}_2[x, y]$ constant along the generic line lies in $\mathbb{F}_2[x{+}y]$: expanding $g(x, x + u) = \sum_i g_i(u)\, x^i$, constancy gives $g_i(\sigma) = 0$ for $i \geq 1$, hence $g_i = 0$, so $g = g_0(x + y)$. □

It remains to prove:

Theorem 12 (no witnesses)**.** *If $Dq \neq 0$, no witness exists.*

4 The Frobenius descent

Throughout, $'$ denotes d/dt and degrees are degrees in t. Over $K[t]$: squares have zero derivative, and conversely a polynomial with zero derivative is a square (it is a polynomial in t^2, and the coefficients have square roots since K is perfect).

Set $w := \gcd(A_1, A_2, A_3)$ and $A_i = wB_i$, so the B_i have no common nonunit factor and

$$q = A_0^2 + w^2\, h, \qquad h := xB_1^2 + yB_2^2 + xyB_3^2, \tag{4}$$

with $B_3 \neq 0$ and partials $h_x = B_1^2 + yB_3^2$, $h_y = B_2^2 + xB_3^2$, $Dh = B_3^2$. We call the gcd extraction (4) the *peel*; it removes the possible curve of critical points of q along $\{w = 0\}$, and Lemma 15 shows none survives in h. Three lemmas, then the descent.

Lemma 13 (no algebraic relation along a witness)**.** *Let (α, β) be a witness with level c, and $f \in \mathbb{F}_2[x, y]$, $f \neq 0$. Then $f(\alpha, \beta) \neq 0$ in $K[t]$.*

Proof sketch. Suppose $f(\alpha, \beta) = 0$; after swapping variables assume α nonconstant. View f and $g := q{+}z$ as polynomials in y over $\mathbb{F}_2[x, z]$: since g is monic and linear in z it is irreducible and cannot divide the z-free f, so they are coprime, and a Bézout identity with denominators cleared gives $f\,U + (q + z)\,V = E(x, z) \neq 0$ in $\mathbb{F}_2[x, z][y]$. Substituting $y \mapsto \beta$, $x \mapsto \alpha$, $z \mapsto c$ kills both summands ($q(\alpha, \beta) + c = 0$), so $E(\alpha, c) = 0$; expanding $E = \sum_i \rho_i(z)\, x^i$ and using that $1, \alpha, \alpha^2, \ldots$ have strictly increasing degrees in t, every $\rho_i(c) = 0$, and some $\rho_i \neq 0$ exhibits c as algebraic over $\mathbb{F}_2$ — contradiction. □

Lemma 14 (algebraic points)**.** *If $u, v \in \mathbb{F}_2[x, y]$ have no common nonunit factor and $(a, b) \in K^2$ satisfies $u(a, b) = v(a, b) = 0$, then a, b are algebraic over $\mathbb{F}_2$.*

Proof sketch. In $\mathbb{F}_2(x)[y]$ a common divisor would descend to $\mathbb{F}_2[x, y]$ by Gauss's lemma [3, Ch. IV], so u, v are coprime there; a cleared Bézout identity gives $su + tv = r(x) \neq 0$, and evaluation at (a, b) gives $r(a) = 0$. Symmetrically for b. □

Lemma 15 (coprimality of the peeled partials)**.** *Let $B_1, B_2, B_3 \in \mathbb{F}_2[x,y]$ have no common nonunit factor, with $B_3 \neq 0$, and set $h_x := B_1^2 + yB_3^2$, $h_y := B_2^2 + xB_3^2$. Then h_x and h_y have no common nonunit factor.*

Proof. Both are nonzero: applying ∂_y to a relation $h_x = 0$ would give $B_3^2 = 0$. Suppose a prime p divides both.

If $p \mid B_3$: subtracting the B_3^2-terms, p divides B_1^2 and B_2^2, hence (primality) all three B_i — excluded.

If $p \nmid B_3$: write $h_x = pu$ and $h_y = pv$. Differentiate, remembering squares are derivative-free:

$$\begin{aligned} 0 = \partial_x h_x = p_x u + p\, u_x \quad &\Longrightarrow \quad p \mid p_x u; \\ B_3^2 = \partial_y h_x = p_y u + p\, u_y \quad &\Longrightarrow \quad p \nmid u \end{aligned}$$

(else $p \mid B_3^2$, so $p \mid B_3$);

$$B_3^2 = \partial_x h_y = p_x v + p\, v_x \quad \Longrightarrow \quad p \nmid p_x$$

(same reason). So the prime p divides $p_x u$ but neither factor — contradiction. □

Proof of Theorem 12. Among all witnesses take one minimizing $n := \deg\alpha + \deg\beta$ (≥ 1, as not both are constant); fix $r := \sqrt{c}$, transcendental (if r were algebraic, so were $c = r^2$). By Lemma 13 applied to the nonzero polynomials w, h_x, h_y, B_3, none of

$$W := w(\alpha,\beta), \quad A := h_x(\alpha,\beta), \quad B := h_y(\alpha,\beta), \quad \Delta := B_3(\alpha,\beta)^2$$

vanishes in $K[t]$.

Peel. From (4) and $q(\alpha,\beta) = c = r^2$, using $(u+v)^2 = u^2 + v^2$,

$$W^2 H = S^2, \qquad H := h(\alpha,\beta), \quad S := r + A_0(\alpha,\beta); \tag{5}$$

differentiating (W^2 and S^2 are squares) gives $W^2 H' = 0$, so $H' = 0$.

Derivative identities. The chain rule for h, with $\partial_x h_x = \partial_y h_y = 0$ and $\partial_y h_x = \partial_x h_y = Dh = B_3^2$, gives

$$A\alpha' + B\beta' = H' = 0, \qquad A' = \Delta\beta', \qquad B' = \Delta\alpha',$$

whence $(AB)' = \Delta(B\beta' + A\alpha') = 0$ and AB is a square in $K[t]$.

Case 1: A and B share a root $\theta \in K$. Then $P := (\alpha(\theta), \beta(\theta))$ is a common zero of h_x, h_y, which by Lemmas 15 and 14 has algebraic coordinates. Evaluating (5) at θ, $S(\theta)^2 = w(P)^2\, h(P)$ is algebraic, hence so is $S(\theta)$, hence so is $r = S(\theta) + A_0(P)$ — contradicting transcendence of r.

Case 2: A, B coprime. Coprime factors of a square are squares (units of $K[t]$ are constants of the algebraically closed K, hence squares), so $A' = B' = 0$; since $\Delta \neq 0$, the derivative identities force $\alpha' = \beta' = 0$, so $\alpha = \alpha_1^2$, $\beta = \beta_1^2$ with $\deg\alpha_1 + \deg\beta_1 = n/2$. Because the coefficients of q lie in $\mathbb{F}_2$, squaring commutes with substitution into q:

$$q(\alpha_1,\beta_1)^2 = q(\alpha_1^2, \beta_1^2) = c,$$

so $q(\alpha_1,\beta_1) = r$ and (α_1,β_1) is a witness of total degree $n/2 < n$ — contradicting minimality. (If $n = 1$ halving is impossible and this branch cannot occur, so the descent is well-founded.) □

Theorem 12 gives (†) by Proposition 11, hence $x + y \notin \mathrm{Clo}(q)$ by Proposition 10 when $Dq \neq 0$; with Proposition 9 this proves Theorem 3, and with Lemma 7, Corollary 5.

5 From $\mathbb{F}_2$ to the dichotomy

Proposition 16. *For every perfect field k of characteristic 2 and every $q \in k[x, y]$, at least one of $x + y$, xy lies outside* $\mathrm{Clo}(q)$*.*

Proof. The descent runs over k with one change: coefficients of q are no longer fixed by squaring, so in the halving branch $q(\alpha_1^2, \beta_1^2) = q^{(1/2)}(\alpha_1, \beta_1)^2$, where $q^{(1/2)}$ applies the inverse Frobenius to each coefficient (perfectness enters here); the halved pair is a witness for $q^{(1/2)}$, and since the induction is on witness degree with q universally quantified inside, the twist is absorbed. The parity decomposition, Proposition 11, and Proposition 10 are unaffected. □

Proof of Theorem 4. k embeds into its perfection $k^{1/2^\infty}$; by Lemma 7, membership of both $x+y$ and xy in $\mathrm{Clo}(q)$ over k would persist over the perfection, contradicting Proposition 16. □

Proof of Theorem 1. If 2 is a unit, $x^2 - y$ generates $R[x, y]$ (Theorem 2). If not, 2 lies in a maximal ideal $\mathfrak{m}$; a generator would push forward along $R \to R/\mathfrak{m}$ (Lemma 7) to one over the residue field, of characteristic 2 — contradicting Theorem 4. □

All results have been machine-verified in Lean 4 against Mathlib (axioms: propositional extensionality, quotient soundness, choice); the development accompanies the paper.

Three questions remain open. Which degrees ≥ 3 admit single generators over $\mathbb{Q}$ ($x^3 - y$ is a candidate; over $\mathbb{F}_3$, $\mathrm{Clo}(x^3 - y)$ consists of additive-plus-constant polynomials, so $x^3 - y$ does not generate there)? Which p generate when 2 is a unit ($xy + 1$ and $x^2 + y$ do not, by a characteristic-zero instance of (†) and by a positivity obstruction, respectively)? Is membership in $\mathrm{Clo}(p)$, or generation, decidable? The identity $q(f, q(f, g)) = g$ for $q = x^2 + y$ over $\mathbb{F}_2$ shows substitution admits unbounded cancellation, so bounded search proves nothing.

References

[1] E. Aichinger, *Generating polynomials using function composition*, Quaest. Math. 36 (2013), no. 1, 39–46.

[2] E. Aichinger and S. Kreinecker, *Subnearrings of* $(\mathbb{Z}[x], +, \circ)$, preprint, arXiv:1709.01345 (2017).

[3] S. Lang, *Algebra*, rev. 3rd ed., Graduate Texts in Mathematics 211, Springer, New York, 2002.

[4] H. M. Sheffer, *A set of five independent postulates for Boolean algebras, with application to logical constants*, Trans. Amer. Math. Soc. 14 (1913), 481–488.

[5] D. L. Webb, *Generation of any n-valued logic by one binary operation*, Proc. Nat. Acad. Sci. USA 21 (1935), 252–254.